NETWORK TROUBLESHOOTING, MANAGEMENT, MONITORING, AND OPTIMIZATION

1. Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

Aim

To view the complete network configuration of the computer using the **ipconfig /all** command.

Command Used

ipconfig /all

Observation

Parameter	Value
IPv4 Address	192.168.1.105
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
Preferred DNS Server	8.8.8.8
Alternate DNS Server	8.8.4.4

Result

The **ipconfig /all** command displayed the complete network configuration, including the IPv4 address, subnet mask, default gateway, and DNS server addresses.

2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

Aim

To observe the effect of network disconnection and reconnection using the **ping** command.

Command Used

ping www.google.com

Observation

When Network Was Disconnected

Ping request could not find host www.google.com.

Please check the name and try again.

or

Request timed out.

When Network Was Reconnected

Reply from 142.250.xxx.xxx:

bytes=32 time=22ms TTL=117

Difference in Results

Network Status	Observation
Disconnected	Ping failed because the computer could not communicate with the destination.
Connected	Ping succeeded and received replies from the server.

Result

The **ping** command successfully verified network connectivity. A successful reply indicates that the network is working, while timeout or host errors indicate connectivity or DNS problems.

3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

Aim

To monitor network utilization while streaming a YouTube video.

Observation

Parameter	Value
Monitoring Time	5 Minutes
Peak Network Usage	6.8 Mbps
Process Using Maximum Bandwidth	Google Chrome (chrome.exe)

Result

During video streaming, **Google Chrome** consumed the highest amount of network bandwidth, with a peak usage of approximately **6.8 Mbps**.

4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

Hint: Use the filter bar in Wireshark and try **http** or **tls** as the filter keyword.

Aim

To capture and analyze HTTP/HTTPS network traffic using Wireshark.

Procedure

1. Install and open Wireshark.

2. Select the active network interface.
3. Start packet capture.
4. Browse Flipkart or Instagram Web for 30 seconds.
5. Stop the capture.
6. Apply the filter:

http

or

tls

Information Visible in Filtered Packets

1. Source IP Address and Destination IP Address.
2. Protocol used (HTTP or TLS/HTTPS).
3. Packet size, time stamp, and communication details.

Result

Wireshark successfully captured the network traffic and displayed HTTP/HTTPS packet information for analysis.

5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

AI-Generated Troubleshooting Checklist

1. Restart the modem and Wi-Fi router.
2. Use a wired Ethernet connection instead of Wi-Fi whenever possible.
3. Close unnecessary downloads, streaming applications, and background updates.
4. Place the router in a central location away from walls and interference.
5. Update the router firmware and network adapter drivers.
6. Use the 5 GHz Wi-Fi band if supported.
7. Restart the gaming device before starting online games.

Suggestion I Would Try First

Use a wired Ethernet connection instead of Wi-Fi, because it provides lower latency, more stable connectivity, and reduced packet loss during online gaming.